Reza Nouralizadeh Ganji

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Publications

Sentiment Analysis of Short and Incomplete Text Published
***Ganji, R.N.**, Tohidi, N.* 2025

- Ganji, R.N. and Tohidi, N., 2025. Sentiment Analysis of Short and Incomplete Texts Using Transformer-Based Attention Models. Journal of Electronic & Information Systems, 7(2), 132–153.

PAMR: Persian Abstract Meaning Representation Corpus Published
*Tohidi, N., Dadkhah, C., **Ganji, R.N.**, Sadr, E.G., Elmi, H.* 2024

- Tohidi, N., Dadkhah, C., Ganji, R.N., Sadr, E.G. and Elmi, H., 2024. PAMR: Persian Abstract Meaning Representation Corpus. ACM Transactions on Asian and Low-Resource Language Information Processing, 23(3), pp.1-20.

Improving Sentiment Classification for Hotel Recommender System Published
***Ganji, R.N.**, Dadkhah, C., Tohidi, N.* 2023

- Ganji, R.N., Dadkhah, C. and Tohidi, N., 2023. Improving Sentiment Classification for Hotel Recommender System through Deep Learning and Data Balancing. Computación y Sistemas, 27(3), pp.811-825.

Research Experience

AI Researcher — Robust NLP & Representation Learning 2022 – 2025
Study: Advanced Sentiment Polarity Detection for Short and Incomplete Texts

- **Motivation:** The presence of short and incomplete text, such as tweets, with misspellings, grammatical errors, and a lack of context could cause critical challenges for sentiment analysis and traditional NLP models.
- **Methodology:** For precise classification, a novel 3-phase system was architected to handle noisy texts. It autocorrects data, uses autoencoder nets for denoising, and fuses intermediate features extracted from RoBERTa LLM.
- **Findings:** The requirement for human annotation is eliminated and SotA results are achieved by attaining F1-scores of 89.96% on Sentiment 140 & 76.91% on ACL 14. The system surpassed previous baselines with a 10% boost in accuracy.

AI Researcher — Computational Linguistics & Semantics 2021 – 2023
Study: Development and Evaluation of the First Persian AMR Corpus

- **Motivation:** The lack of essential semantic resources, like the AMR corpus for low-resource languages like Persian, can restrict advanced NLP research like semantic parsing and text generation.
- **Methodology:** To develop the AMR corpus, 1,020 sentences were annotated by modifying guidelines for unique Persian features. The data augmentation was performed to generate 888 synthetic sentences from the corpus.
- **Findings:** Co-developed and released the first Persian AMR corpus. Its use in data augmentation boosted a sentiment analysis model's F1-score and accuracy by 12%.

AI Researcher — Deep Learning & Recommender Systems 2021 – 2023
Study: Enhancing Hotel RS with Deep Learning and Data Balancing

- **Motivation:** Sentiment-driven hotel recommenders exhibit bias due to imbalanced data (an excessive number of positive reviews) and multilingual text, which degrades classification accuracy.
- **Methodology:** For precise recommendation, an RS was proposed to benefit from sentiment analysis. It utilized a T5 model for data balancing via augmentation, and a cross-lingual XLM-RoBERTa classifier with an attention mechanism used model's hidden states.
- **Findings:** It achieved an 89% F1-score on the TripAdvisor data and surpassed baselines by 5%. Its efficient integrated architecture cuts inference time by over 60% compared to the baseline.

Education

Master of *Artificial Intelligence* 2020 – 2023
K. N. Toosi University of Technology Tehran, Iran

- ***Notable Courses:*** Natural Language Processing, Neural Networks, Recommender Systems, Information Retrieval, Evolutionary Computation
- ***Thesis:*** Sentiment Analysis of Short and Incomplete Text using Transformers and Attention Mechanism; under supervision of *Dr. Chitra Dadkhah*
- ***Thesis Grade:*** (20/20 – 4/4) & ***GPA:*** (18.32/20 – 3.88/4)

Bachelor of *Computer (Software) Engineering* 2017 – 2020
Shomal University Amol, Iran

- ***Notable Courses:*** Machine Learning, Artificial Intelligence, Algorithm Design, Data Structures, Formal Languages and Automata Theory, Engineering Probability and Statistics
- ***Thesis:*** A machine learning-based model for spam detection on mobile phone short message service (SMS); under supervision of *Dr. Hamidreza Koohi*
- ***Thesis Grade:*** (20/20 – 4/4) & ***GPA:*** (17.61/20 – 3.44/4)

Research Interests

Natural Language Processing | Deep Learning | Machine Learning
Information Retrieval | Large Language Models | Mechanistic Interpretability

Technical Skills

Programming: Python
Deep Learning: Transformers, Attention mechanisms, Large Language Models (LLMs), Recurrent Neural Network (RNN), Long Short Term Memory (LSTM), Gated Recurrent Unit (GRU), Autoencoders
Machine Learning: Clustering, Decision Tree, Support Vector Machine (SVM), Multi-Layer Perceptron (MLP), Ensemble Models, Logistic Regression
Math/Theory: Linear Algebra, Probability & Statistics, Multivariate Calculus, Optimization Methods
AI Packages: PyTorch, Hugging Face, Numpy, Pandas, Matplotlib, WandB, PLotly, Scikit-learn
Languages: Persian (Native), English (C1)

Licenses & certifications

TOEFL iBT ETS
Overall Score: 100/120 (5.0/6) — CEFR Level C1 July 2026

Natural Language Processing Specialization Coursera
Younes Bensouda Mourri, Łukasz Kaiser February 2022

- In this four-course specialization, students learn how to construct applications for NLP activities including question answering and sentiment analysis, and how to create translation, summarization, and chatbot tools.

Deep Learning Specialization Coursera
Andrew NG, Kian Katanforoosh, Younes Bensouda Mourri December 2021

- The five courses in this specialization educate students how to design, develop, and optimise CNNs, RNNs, LSTMs, and Transformers utilising Dropout, BatchNorm, Xavier/He initialization, and other approaches.

Conferences & Presentations

Neural-based approaches for sentiment analysis February 2022
K.N.T. University Master's Research Seminar

Applications of Monte Carlo sampling in data mining June 2021
K.N.T. University Data Mining's Research Seminar

Bio-Inspired algorithms for sentiment analysis May 2021
K.N.T. University Evolutionary Computation's Research Seminar

How do search engines use machine learning methods? May 2019
Shomal University Artificial Intelligence's Research Seminar